

Synthesis — what the four territories share and why they differ

This note distils the cross-case argument of chapter 11, tying it back to the per-territory numbers in [RESULTS.md](#) and the method in [METHODOLOGY.md](#).

1. The longitudinal argument

Read coast-to-west, the four territories trace one line: **the later the critical juncture arrived relative to the frontier, the more forest the policy was able to keep**. The geographic order is also a policy-history order and a "March to the West" frontier order.

Territory	Biome	Critical juncture	Title	Extent	Forest	Deforest.	One-line story
Krenak	Atlantic Forest	1808	1997	20,034 ha	~11%	~89%	Recovery on exhausted land
Bacurizinho	Cerrado–Amazon	1901	1983 / 2008	135,346 ha	~78%	~12%	Boundary-effect experiment
Kayapó	Amazon	1970s	1991	3,283,191 ha	~93%	~5%	Extractive lockstep, intact core
Betânia	Amazon	1960s	1995	126,066 ha	~93%	~2%	Hydrological re-classification

A territory that lost protection early and spent a century under extractive tenure shows ~89% deforestation; one that entered the post-Constitution regime nearly intact shows 2–5%. Biome contributes (the Atlantic Forest was already fragmented before the satellite era) but **biome alone cannot explain an 87 pp gap** — the buffer contrast (§4) shows unprotected surroundings evolved very differently under the same biome.

2. What each territory teaches

- **Krenak — recognition without resource.** Policy recognition is necessary but insufficient once path dependence has exhausted what the policy protects. The 1997 STF ruling returned the title, not the forest; the post-1988 story is slow recovery of degraded land, plus a non-policy shock (Fundão 2015) the pipeline must accommodate.
- **Bacurizinho — tenure completeness.** The *same* legal category yields different pixel outcomes when tenure clarity differs. The 1.75–1.91× 4901-vs-4902 differential is the cleanest internal proof that physical demarcation and resolution of overlapping CAR claims — not the legal label — is what bites.
- **Kayapó — institutional capacity.** Three million hectares held above 91% cover against an aggressive frontier is not natural; it is warrior strategy, the Altamira Gathering, Instituto Raoni/Kabu, guard posts.

But the 2019 → 2020 loss uptick shows even that is not autonomous from national politics, and the mining-pasture lockstep is the live pressure front.

- **Betânia — the upper bound.** Where stewardship and recent recognition run cleanly, the record shows what an effective regime looks like: >92% cover, a confirmed *null* result for COVID-2020, and a dominant signal that is hydrological, not agricultural.

3. Policy lag has three temporal signatures

The framework's "lag" claim survives the data — but the lag is **not a single number**. The Yvynation outputs let three distinct kinds be told apart:

Signature	Definition	Cleanest examples
(a) Fast lag (1–2 yr)	A discrete national event → next-year Hansen canopy hit	Krenak 2012 Code → 2014 loss; Kayapó 2016 PEC215/Belo Monte → 2017 (73,644 ha); Kayapó 2019 rollback → 2020 (32,074 ha)
(b) Persistent / boundary effect	Visible in the 2001–2023 loss percentage , not any single year	Bacurizinho 4901 vs 4902 (1.91×); Kayapó mining-pasture lockstep (+1,352% each)
(c) Gradual re-classification	Multi-year MapBiomas class drift no single year can carry	Krenak forest recovery (+141%); Betânia Floodable Forest → Wetland (+223%)

If the lag were zero, policy would have to anticipate satellite coverage; if it were infinite/random, path dependence would carry no explanatory content. The finite, patterned lag is exactly the "policy-to-pixel signature" the framework was built to capture — and separating the three signatures is one of the project's main methodological results.

4. The buffer protective effect is real and quantifiable

Using the 10 km exclusive buffer as a counterfactual (see METHODOLOGY §5), the May 2026 Collection 10 batch ([multi_window.md](#)) shows the interior–buffer divergence directly, and it is consistent:

Region pair	Territory forest Δ	Buffer forest Δ
Bacurizinho 4901	–1.3 pp	–25.5 pp
Bacurizinho 4902	–6.2 pp	–22.0 pp
Kayapó NE (hot spot)	–0.9 pp	–65.2 pp (91% → 26% forest)
Kayapó NW	–0.2 pp	–21.7 pp
Kayapó SE	–1.2 pp	–31.3 pp
Betânia	–1.6 pp	–1.4 pp (near-identical)
Krenak Sete Salões	+6.4 pp (regrowth)	+0.2 pp

The pattern maps onto the gradient: the contrast is sharpest where the frontier is hottest (Kayapó NE/SE, Bacurizinho), *absent* where the surroundings are still deep forest (Betânia), and *inverted* where the territory is recovering inside an already-cleared matrix (Krenak). The sharp satellite-visible line at the boundary means enforcement works at the limit — and that the actors clearing land plan around that line using the same spatial data these analyses use.

5. Same national laws, different outcomes — four structural conditions

The 1965 Code, 1988 Constitution, 2004 PPCDA and 2012 Code appear in all four policy-lag tables, yet bite differently. Four conditions determine whether a national rule has a local effect:

1. **Path-dependence depth** — how exhausted was the resource when the rule arrived? (Krenak: nothing left to protect; Betânia: everything intact.)
2. **Tenure completeness** — was declaration backed by physical demarcation and resolution of private claims? (Bacurizinho's expansion is the counter-example.)
3. **Institutional capacity on the ground** — organisation, resources, visibility to enforce the boundary. (Kayapó.)
4. **Biome & market proximity** — recovery ceilings and propagule pressure differ between a fragmented Atlantic Forest matrix and intact Amazon.

These are not simply additive: the **absence of any one can defeat the others**. Bacurizinho has capacity and an intact biome, but incomplete tenure let its expansion deforest at buffer-like rates. Path dependence is the strongest single predictor — *but not destiny*: Kayapó and Betânia show that early recognition plus institutional response can reset a trajectory.

6. Comparative valuation — the economic stakes of the gradient

Territory	Land market value (USD)	Annual ES value (USD)	ES as % of land value
Krenak	\$41.9 M	\$8.2 M	19.6%
Bacurizinho	\$211.5 M	\$55.5 M	26.2%
Kayapó	\$7,685.8 M	\$1,346.1 M	17.5%
Betânia	\$19.5 M	\$51.7 M	265.0%

Two structural facts fall out of the gradient:

- **The Betânia inversion.** Remote deep-forest land has the lowest per-hectare market price ("General" regional typology) but the same ~410 USD/ha/yr regulating-service value as anywhere — so its ecosystem-service value is 2.6× its market value. The opposite logic hits Krenak: urban proximity inflates land price while 89% deforestation suppresses ES potential, making it the territory where the opportunity cost of conversion was most fully realised and the recovery cost is now highest.
- **Kayapó's magnitude.** ~\$1.3 B/yr in ecosystem services (~\$410/ha × 3.28 M ha) dwarfs the other three combined — and still sits below the ~797 USD/ha/yr opportunity cost of agricultural conversion, which

is precisely why the surrounding frontier keeps pushing and the institutional network must keep absorbing the pressure.

7. Bottom line for the thesis

Path dependence explains both the **persistence of degradation** (Krenak) and the **possibility of trajectory change** (Kayapó, Betânia). The critical-juncture concept does not predict equilibrium; it predicts that the cost of changing a trajectory rises with its depth — so **early intervention is more efficient than late recognition**. The Yvynation batch is the instrument that makes this measurable and reproducible for any territory in the FUNAI (or CNUC) database.

Open threads flagged in the report

- Reconcile the Collection 9 chapter figures with the Collection 10 batch on the Collection 11 release (August 2026), then extend Hansen to 2026 in early 2027.
- Run the full four-territory 10 km exclusive-buffer batch (partly done; Kayapó quadrant buffers already in [multi_window.md](#)).
- Add the CAR-overlap and 2012/2017 fire-scar overlays for Bacurizinho.
- Integrate the planned policy/political timeline layer (legislation + presidents /governors) into the deforestation-timeline chart for tendency analysis.